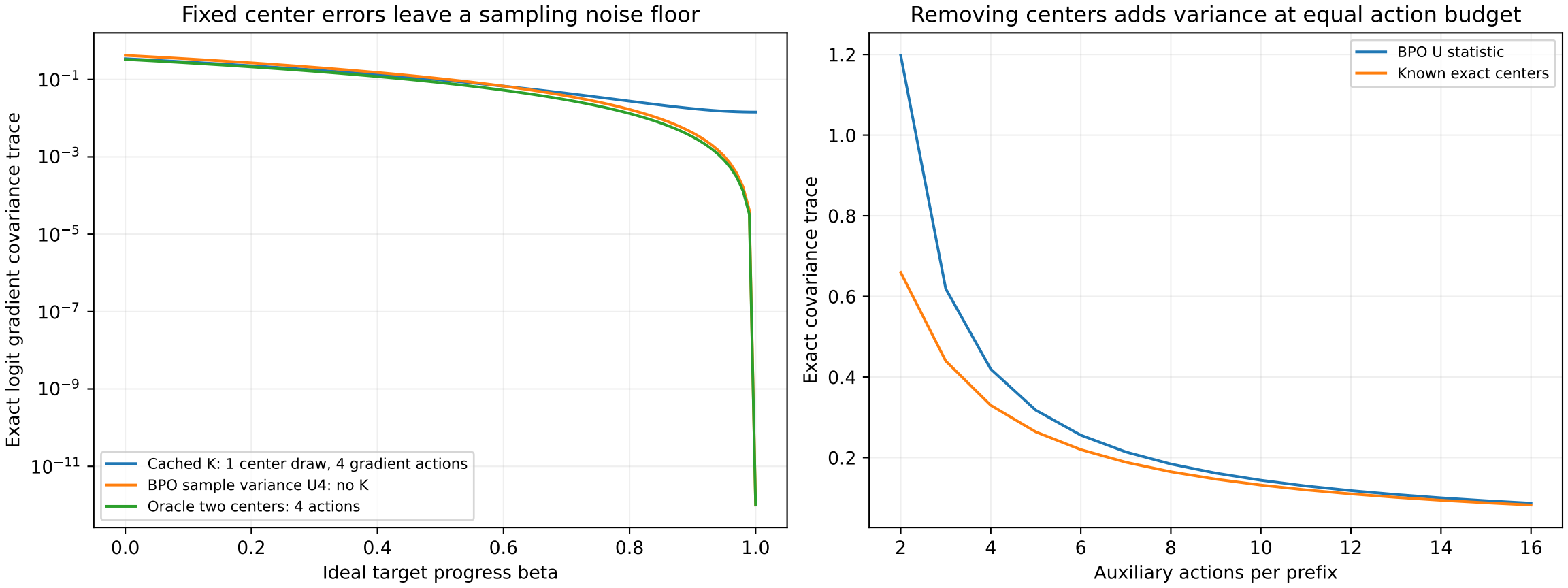
Fixed center errors leave a sampling noise floor
Exact logit gradient covariance trace
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
Cached K: 1 center draw, 4 gradient actions
BPO sample variance U4: no K
Oracle two centers: 4 actions
0.0
0.2
0.4
0.6
0.8
1.0
Ideal target progress beta
Removing centers adds variance at equal action budget
Exact covariance trace
1.2
1.0
0.8
0.6
0.4
0.2
BPO U statistic
Known exact centers
2
4
6
8
10
12
14
16
Auxiliary actions per prefix